VanDrishti

AI-Based Forest Health Monitoring System

Product Requirements Document | Hackathon 2026

Version	1.0 — Final	**Date**	March 18–19, 2026
Team	4 members (backend x2, frontend x1, ML x1)	**Duration**	48 hours
Hardware	Raspberry Pi 4 + PCs	**Status**	In Development

1. Executive Summary

VanDrishti *(Hindi: Forest Guardian)* is an AI-powered forest health monitoring system that detects slow, invisible forest degradation weeks before it becomes visible to the human eye. Unlike existing solutions focused on fire detection, VanDrishti monitors the silent killers of forest ecosystems: canopy thinning, biodiversity collapse, soil stress, and land-use change.

The system fuses free satellite imagery (Sentinel-2), biodiversity signals (GBIF / eBird), thermal anomaly data (NASA FIRMS), and real-time climate feeds (OpenWeatherMap) into a single composite Forest Health Index (FHI) score per zone, updated continuously. A Raspberry Pi 4 acts as an on-site edge intelligence node — polling live data APIs, computing a local micro-FHI score, buffering offline, and syncing to the cloud dashboard over LAN.

2. Problem Statement

2.1 The Silent Killer Problem

Current forest monitoring is reactive. Systems like MODIS or VIIRS detect fires — but a forest does not die in a fire. It dies slowly over weeks through:

- Canopy thinning and crown die-back
- Declining biodiversity as indicator species disappear
- Soil moisture stress from irregular rainfall patterns
- Illegal logging creating small but cumulative clearings
- Land encroachment at forest boundaries

By the time rangers observe these signs on foot, the ecological tipping point has often already passed.

2.2 Gaps in Existing Solutions

Existing Solution	What It Does	Critical Gap
MODIS / VIIRS (NASA)	Detects active fire pixels	No early-stage degradation detection
Global Forest Watch	Annual Hansen deforestation data	5-day to yearly cadence — too slow
Manual ranger patrol	Ground-level observation	Covers <5% of forest area per day
IoT sensor networks	In-situ temperature, humidity	Expensive, require infrastructure
VanDrishti (Ours)	Multi-signal AI degradation index	Fills all gaps above

3. Product Vision & Goals

3.1 Vision Statement

"Make the invisible visible — detect forest death in slow motion before it becomes irreversible."

3.2 Goals for the 48-Hour Hackathon

- **G1** Deliver a working end-to-end system: satellite data ingestion → AI processing → live dashboard
- **G2** Demonstrate Raspberry Pi as a deployable edge node with live API polling and heartbeat sync
- **G3** Show 30-day historical degradation trends for real Indian forest zones (Corbett / Sundarbans)
- **G4** Compute and display the Forest Health Index (FHI) in real time with zone-level alerts
- **G5** Demonstrate carbon stock estimation as a monetisable output

4. System Architecture

4.1 Device Roles

Device	Role	Runs On	Connects To
Team PC (Server)	Main backend, AI engine, dashboard host	Python FastAPI + React	Internet + RPi via LAN
Raspberry Pi 4	Edge intelligence node, live API poller	Python 3 + SQLite	Team PC via Wi-Fi LAN

4.2 Architecture Layers

Layer 1 — Data Ingestion (PC)

- Pull 2 months of Sentinel-2 NDVI band data for Corbett and Sundarbans via Copernicus Dataspace API
- Ingest NASA FIRMS thermal anomaly CSV feed — updated every 3 hours
- Fetch GBIF species observation data per zone coordinates
- Pull eBird recent observations as biodiversity health proxy
- OpenWeatherMap: rainfall, temperature, humidity per zone

Layer 2 — AI / ML Processing (PC)

- NDVI Change Detector: pixel-level diff between bi-weekly Sentinel-2 images using rasterio + numpy
- Biodiversity Anomaly Scorer: Isolation Forest on GBIF species count time-series
- Moisture Stress Index: derived from rainfall deficit and temperature anomaly
- Land-Use Change Detector: OpenCV-based canopy gap ratio from NDVI raster
- Risk Predictor: 7-day tipping point forecast using linear regression on FHI trend slope

Layer 3 — Forest Health Index Aggregation (PC)

Input Signal	Weight	Data Source	Update Frequency
NDVI canopy change	35%	Sentinel-2 (Copernicus)	Every 5 days
Biodiversity signal	25%	GBIF + eBird	Daily
Thermal anomaly count	20%	NASA FIRMS	Every 3 hours
Moisture stress	15%	OpenWeatherMap	Hourly
RPi edge micro-FHI	5% (bonus)	Raspberry Pi agent	Every 5 minutes

FHI = weighted sum of all signals, normalised to 0–100. Score below 50 triggers yellow alert. Score below 30 triggers red alert.

Layer 4 — Raspberry Pi Edge Node

- Polls NASA FIRMS, OpenWeatherMap, and GBIF APIs every 5 minutes
- Computes local micro-FHI using lightweight rule-based engine (no GPU required)
- Sends JSON heartbeat to PC backend via HTTP POST over LAN
- If LAN connection drops: stores readings in local SQLite database, syncs on reconnect
- Displays live status on terminal (visible to judges as physical proof of hardware)

5. Features & Requirements

5.1 Core Features (Must Ship by Hour 36)

Feature ID	Feature Name	Description	Priority
F01	Satellite NDVI Ingestion	Download and process 2-month Sentinel-2 NDVI rasters for 2 forest zones	P0
F02	FHI Computation Engine	Weighted multi-signal composite score per zone, updated on new data	P0
F03	Live Map Dashboard	Leaflet.js satellite map with colour-coded zone health overlay (red/yellow/green)	P0
F04	Alert System	Real-time alert panel: zone ID, FHI score, change delta, threat type	P0
F05	30-Day Trend Graph	Recharts line graph showing FHI over time per selected zone	P0
F06	RPi Edge Agent	Python poller running on RPi, sending live heartbeats to PC dashboard	P0
F07	Offline Buffer	RPi SQLite buffer — stores data when LAN drops, auto-syncs on reconnect	P1
F08	7-Day Risk Forecast	ML-predicted FHI trend line — flags zones approaching tipping point	P1

5.2 Bonus Features (Ship if Time Permits)

Feature ID	Feature Name	Description
F09	Carbon Stock Estimator	Derive above-ground biomass from NDVI and display as carbon tonnes per zone
F10	Degradation Heatmap	Pixel-level change intensity overlay on satellite basemap
F11	Zone Comparison View	Side-by-side FHI trend comparison for two zones
F12	Ranger Alert Export	One-click PDF report for a zone — FHI summary + trend + alert log

6. APIs & Data Sources

API / Source	What We Use It For	Auth	Rate Limit / Cost
Copernicus Dataspace (ESA)	Sentinel-2 NDVI satellite imagery — 2-month download for Corbett + Sundarbans	Free account	Free tier sufficient
NASA FIRMS API	Real-time thermal anomaly points — fires, heat stress, burning	Free API key (instant)	Unlimited for research
GBIF REST API	Species observation counts per coordinate bounding box	No auth needed	Free, 1M calls/day
eBird API 2.0 (Cornell)	Recent bird observation counts per region — biodiversity proxy	Free API key	Free for non-commercial
OpenWeatherMap API	Temperature, humidity, rainfall per zone — moisture stress	Free API key	1000 calls/day free
Global Forest Watch API	Hansen annual tree cover loss data — baseline comparison	No auth	Free
Nominatim (OpenStreetMap)	Reverse geocoding zone coordinates to human-readable names	No auth	1 req/sec free

Note: All Sentinel-2 data will be pre-downloaded before the hackathon begins (2 months of data for 2 zones = approximately 800 MB). All other APIs are polled live during the demo.

7. Technology Stack

7.1 Backend (Team PC)

Component	Technology	Purpose
API Server	FastAPI (Python 3.11)	REST endpoints + WebSocket for live dashboard updates
Satellite Processing	rasterio, numpy, scipy	NDVI raster parsing, band arithmetic, change detection
Machine Learning	scikit-learn	Isolation Forest anomaly scorer, linear regression risk predictor
Database	PostgreSQL / SQLite	Zone data, FHI history, alert log
Task Scheduler	APScheduler	Periodic API polling and FHI recalculation
Messaging	WebSocket (FastAPI native)	Push live updates to frontend dashboard

7.2 Frontend (Team PC)

Component	Technology	Purpose
Framework	React 18	Component-based UI, state management
Map Layer	Leaflet.js + React-Leaflet	Interactive satellite map with zone polygon overlays
Charts	Recharts	FHI trend graphs, signal contribution bar charts
Styling	Tailwind CSS	Responsive layout, colour-coded alert states
Real-time	WebSocket client	Receives live FHI updates and alerts from backend

7.3 Raspberry Pi Edge Node

Component	Technology	Purpose
Language	Python 3.11	Main agent script
HTTP Client	requests	API polling for FIRMS, GBIF, OpenWeatherMap
Local Storage	SQLite3 (built-in)	Offline data buffer when LAN drops
Sync Protocol	HTTP POST over LAN	Heartbeat to PC backend every 5 minutes
Display	Rich (Python CLI)	Terminal dashboard showing live readings — visible to judges

8. 48-Hour Work Plan

8.1 Team Assignments

Person	Role	Hours 0–12	Hours 12–30	Hours 30–48
Person 1	Backend Lead	FastAPI setup, DB schema, Sentinel-2 parser	FHI engine, WebSocket server, API integrations	Integration, bug fixes, demo rehearsal
Person 2	RPi Engineer	RPi setup, FIRMS + weather poller running	Offline buffer, heartbeat sync to PC, micro-FHI	Polish terminal display, ensure stable LAN link
Person 3	Frontend	React scaffold, Leaflet map with zone polygons	FHI charts, alert panel, live WebSocket updates	UI polish, mobile view, presentation mode
Person 4	ML Engineer	NDVI change detector, biodiversity scorer	Risk predictor (7-day), carbon estimator	Model tuning, historical data validation

8.2 Hour-by-Hour Milestones

Milestone	Target Time	Definition of Done
M0 — Environment ready	Hour 2	All repos cloned, APIs keys obtained, RPi online, Sentinel-2 data downloading

Milestone	Target Time	Definition of Done
M1 — Data pipeline live	Hour 8	Sentinel-2 NDVI parsed into numpy arrays; FIRMS + weather data flowing into DB
M2 — FHI calculating	Hour 14	FHI score computed for both zones; RPi sending heartbeat to PC
M3 — Dashboard live	Hour 22	Map showing zones; FHI scores visible; alert panel firing on low FHI
M4 — Full integration	Hour 30	All signals wired; 30-day trend graph working; RPi visible in Connected Nodes panel
M5 — Bonus features	Hour 40	Carbon estimator + 7-day risk forecast + degradation heatmap
M6 — Demo ready	Hour 46	Full run-through rehearsed; RPi on table; pitch deck done

9. Dashboard Specification

9.1 Main Panels

- Satellite Map (full-screen left panel): Leaflet with Sentinel-2 basemap, forest zone polygons colour-coded by FHI (red < 30, amber 30–59, green 60+), click zone to see detail
- FHI Score Card (top right): large numeric FHI score for selected zone, delta vs last week, status badge
- 30-Day Trend Chart (mid right): Recharts line chart showing FHI history, threshold lines at 60 and 30
- Signal Breakdown (mid right): horizontal bar chart showing NDVI, biodiversity, thermal, moisture contributions
- Alert Feed (bottom right): timestamped alert log, zone ID, threat type, severity
- Connected Nodes Panel (bottom bar): shows RPi heartbeat status, last sync time, micro-FHI value

9.2 Alert Severity Definitions

Level	FHI Score	Colour	Meaning	Action
Healthy	60–100	Green	Normal ecosystem activity	Monitor only

Level	FHI Score	Colour	Meaning	Action
Watch	40–59	Amber	Declining trend detected	Ranger awareness
Alert	20–39	Orange	Significant degradation	Ranger deployment
Critical	0–19	Red	Tipping point imminent	Immediate intervention

10. Hardware Demo Plan (Raspberry Pi)

10.1 Physical Setup

- Raspberry Pi 4 placed on the demo table — visible to judges
- Connected to team PC via Wi-Fi hotspot (same LAN, no internet required for demo)
- Terminal window open on RPi showing Rich CLI dashboard: zone name, last reading, micro-FHI, sync status
- LED activity light confirms the device is running (no extra hardware needed)

10.2 Demo Trigger Script

To wow judges in the live demo — manually push a low FHI event:

```
  curl -X POST http://localhost:8000/api/simulate-alert \     -d '{"zone": "corbett_A",
"fhi": 22, "reason": "canopy_loss"}'
```

This fires a red alert on the dashboard, triggers the alert sound, and the RPi terminal reacts within 5 seconds — demonstrating the closed loop in real time.

11. Pitch & Judging Strategy

11.1 One-Line Pitch

"Satellites watch. AI thinks. Edge hardware confirms. Rangers act — before it's too late."

11.2 60-Second Pitch Script

"A forest doesn't die in a fire. It dies slowly — over weeks — through canopy thinning, soil stress, and biodiversity collapse. By the time rangers see it on foot, the tipping point has already passed.

VanDrishti is the world's first composite forest health index — a continuous 0-to-100 score per forest zone, powered by satellite spectral analysis, live biodiversity signals, thermal data, and climate feeds. It predicts ecosystem collapse 7 days before it happens.

We're not detecting fire. We're detecting forest death in slow motion. And we've deployed it on a ₹4000 Raspberry Pi that any forest department in India can install today — zero sensors, just power and a SIM card.

This is ecosystem ICU monitoring. This is VanDrishti."

11.3 Judging Criteria Mapping

Judging Criterion	How We Address It
Innovation / Uniqueness	No existing system combines NDVI + biodiversity + thermal + climate into one predictive FHI score
Technical Depth	Multi-source satellite processing, ML anomaly detection, edge computing on RPi, WebSocket real-time updates
Real-World Impact	Targets India's 21% forest cover; integrates with forest dept workflows; carbon market monetisation path
Working Demo	Live map, live FHI updates, RPi on table sending heartbeats, simulated alert trigger
Hardware Integration	Raspberry Pi 4 as physical edge node — visible, running, demonstrable in real time
Scalability	Stateless API design; any number of RPi nodes can register; satellite zones are config-driven

12. Risks & Mitigations

Risk	Likelihood	Mitigation
Sentinel-2 download too slow during hackathon	High	Pre-download 2 months of data before Day 1 starts
GBIF / eBird returns sparse data for Indian zones	Medium	Use mock biodiversity data as fallback; note limitation in pitch
RPi WiFi drops during demo	Low	Demo with RPi pre-buffered; SQLite sync replays data on reconnect
ML model not converging in time	Medium	Ship rule-based FHI first; ML model is enhancement not core

Risk	Likelihood	Mitigation
Frontend too complex to finish	Medium	Use pre-built Leaflet template; charts are last to add
Time overrun on satellite processing	Medium	Pre-process NDVI into CSVs offline; load from file during demo

13. Success Metrics

- FHI score is live and updating for at least 2 forest zones during the demo
- RPi is physically on the demo table, terminal running, sending visible heartbeats
- Alert fires correctly when FHI drops below threshold — live in front of judges
- 30-day trend graph shows real Sentinel-2 NDVI data — not mock data
- Pitch delivered in under 90 seconds — judges can repeat the one-line pitch back to us

Appendix — Quick Setup Checklist

Before hackathon Day 1 (do tonight):

- Create Copernicus Dataspace account and download Sentinel-2 L2A data for Corbett National Park (bounding box: 78.5°E–79.5°E, 29.0°N–30.0°N) for Jan–Mar 2026
- Sign up for NASA FIRMS API key at firms.modaps.eosdis.nasa.gov
- Sign up for OpenWeatherMap free tier at openweathermap.org/api
- Sign up for eBird API at ebird.org/api/keygen
- Flash Raspberry Pi OS Lite on RPi, enable SSH, set up Wi-Fi, install Python 3.11 and packages: requests, sqlite3, rich, APScheduler
- Create GitHub monorepo with /backend, /frontend, /rpi-agent folders — all 4 team members push from day 1
- Agree on zone naming convention: zone IDs like corbett_A, corbett_B, sundarbans_A

On Day 1 Hour 0:

- Backend Person 1: git clone, pip install, start FastAPI server, confirm /health endpoint returns 200
- RPi Person 2: SSH into RPi, run rpi-agent/setup.sh, confirm heartbeat POST logs appear on PC
- Frontend Person 3: npx create-react-app, install react-leaflet recharts, render a blank map on localhost:3000
- ML Person 4: open Jupyter, load pre-downloaded Sentinel-2 NDVI raster, compute first NDVI change delta

Good luck — go build something that matters. 🌿